



## APPENDIX D: FLIGHT READINESS REVIEW CHECKLIST

Table 8: Flight Readiness Review checklist.

| SECTION                                    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ACTIONS TO BE TAKEN                                   |
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| <b>PROPULSION SYSTEMS</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |
| Checklist                                  | Upon request, the flier should provide the inspector with hardcopy checklist procedures for the propulsion system's safe handling, assembly, disassembly, and operation (both nominal and off-nominal/contingency flows) – including self-inspection/verification steps which make individual team members accountable to one another for having completed the preceding process(es).                                                                                                                                                                                                         | Simple confirmation<br>Inspection on site             |
| Non-toxic Propellants                      | Launch vehicles entering EuRoC shall use non-toxic propellants. Ammonium perchlorate composite propellant (APCP), potassium nitrate and sugar (also known as "rocket candy"), nitrous oxide, liquid oxygen (LOX), hydrogen peroxide, kerosene, propane, alcohol, and similar substances, are all considered non-toxic. Toxic propellants are defined as those requiring breathing apparatus, unique storage and transport infrastructure, extensive personal protective equipment (PPE), etc. Home-made propellant mixtures containing any fraction of toxic propellants are also prohibited. | Simple confirmation                                   |
| Total Impulse                              | The sum of all rocket stages' impulse must either not exceed 40,960 newton-seconds, or the Flier must have previously consulted with EuRoC on provisions for launching a larger rocket.                                                                                                                                                                                                                                                                                                                                                                                                       | Simple confirmation                                   |
| Motor Retention                            | The design must provide for positive retention of the propulsion system within the airframe - leaving no possibility for the propulsion system to shift from its retaining device(s) and jettison itself.                                                                                                                                                                                                                                                                                                                                                                                     | Inspection on site<br>Proof by reasoned argumentation |
| Thrust Structure                           | A "structural chain" that transfers the propulsion system thrust to various points on the rocket structure must exist and it must be capable of withstand these loads.                                                                                                                                                                                                                                                                                                                                                                                                                        | Inspection on site<br>Proof by reasoned argumentation |
| Thrust Curve                               | Upon request, the flier must provide the inspector with hardcopy thrust curve data for each individual rocket motor or engine implemented.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proof by calculation                                  |
| <b>PROPULSION SYSTEM SAFING AND ARMING</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |



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| Pre-flight and Countdown Procedure       | Upon request, the flier should provide the inspector with hardcopy checklist procedures for any of the propulsion system's unique final on-pad preparations, pre-flight, and launch (both nominal and off-nominal/abort/mishap flows) - including self-inspection/verification steps which make individual team members accountable to one another for having completed the preceding process(es).                                                                                                                                                                                                                                               | Simple confirmation<br>Inspection on site                |
| Ground-start Ignition Circuit Arming     | All ground-started propulsion system ignition circuits/sequences shall not be "armed" until all personnel are at least 15 m away from the launch vehicle. The provided launch control system satisfies this requirement by implementing a removable "safety jumper" in series with the pad relay box's power supply. The removal of this single jumper prevents firing current from being sent to any of the launch rails associated with that pad relay box. Furthermore, access to the socket allowing insertion of the jumper is controlled via multiple physical locks to ensure that all parties have positive control of their own safety. | Simple check                                             |
| Air-start Ignition Circuit Arming        | All upper stage (i.e., air-start) propulsion systems shall be armed by launch detection (e.g., accelerometers, zero separation force [ZSF] electrical shunt connections, break-wires, or other similar methods). Regardless of implementation, this arming function will prevent the upper stage from arming in the event of a misfire.                                                                                                                                                                                                                                                                                                          | Proof by reasoned<br>argumentation<br>Inspection on site |
| Propellant Offloading After Launch Abort | Hybrid and liquid propulsion systems shall implement a means for remotely controlled venting or offloading of all liquid and gaseous propellants in the event of a launch abort.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Proof by reasoned<br>argumentation                       |
| Air-start Ignition Circuit Electronics   | All upper stage ignition systems shall comply with same requirements and goals for "redundant electronics" and "safety critical wiring" as recovery systems — understanding that in this case "initiation" refers to upper stage ignition rather than a recovery event.                                                                                                                                                                                                                                                                                                                                                                          | Simple confirmation<br>Inspection on site                |
| Staging Ignition Commit Criteria         | The electronics controlling the various staging events must inhibit staging if the rockets' flight profile deviates from predicted nominal behaviour.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Proof by reasoned<br>argumentation                       |



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| Positive State Indication                           | Each independent set of electronics controlling staging events must provide sensory (i.e., visual or auditory) indication of its activation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Simple confirmation<br>Inspection on site |
| Special Consideration for "Drag Separation"         | The electronics controlling stage ignition in design's implementing "drag-separation" must not be located in the separating stage - where premature separation could prevent ignition of the following stage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Simple confirmation<br>Inspection on site |
| <b>SRAD PROPULSION SYSTEM TESTING</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |
| Combustion Chamber Pressure testing                 | SRAD and modified COTS propulsion system combustion chambers shall be designed and tested according to the SRAD pressure vessel requirements defined in Section 4.2. . Note that combustion chambers are exempted from the requirement for a relief device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Proof by previous testing                 |
| Hybrid and Liquid Propulsion System Tanking Testing | SRAD and modified COTS propulsion systems using liquid propellant(s) shall successfully (without significant anomalies) have completed a propellant loading and off-loading test in "launch-configuration", prior to the rocket being brought to the competition. This test may be conducted using either actual propellant(s) or suitable proxy fluids, with the test results to be considered a mandatory deliverable and an annex to the Technical Report, in the form of a loading and off-loading checklist, complete with dates, signatures (at least three) and a statement of a successful test. Failure to deliver this annex will automatically result in a "denied" flight status. Loading and unloading of liquid propellants must be a well-drilled, safe and efficient operation at the competition launch rails. | Proof by previous testing                 |
| Static Hot-fire testing                             | SRAD propulsion systems shall successfully (without significant anomalies) complete an instrumented (chamber pressure and/or thrust), full scale (including system working time) static hot-fire test prior to EuRoC. In the case of solid rocket motors, this test needs not to be performed with the same motor casing and/or nozzle components intended for use at the EuRoC (i.e., teams must verify their casing design, but are not forced to design reloadable/reusable motor cases). The test results and a statement of a successful test, complete with dates and signatures (at least three) are considered a mandatory deliverable and an annex to the                                                                                                                                                              | Proof by previous testing                 |



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|                                            | Technical Report. Failure to deliver this annex will automatically result in a “denied” flight status. See Section 2.6.6. for more information.                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                 |
| <b>RECOVERY SYSTEMS AND AVIONICS</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                 |
| Checklist                                  | Upon request, the flier must provide the inspector with hardcopy checklist procedures for the recovery system's safe handling, assembly, disassembly, and operation (both nominal and off-nominal/contingency flows) - including self-inspection/verification steps which make individual team members accountable to one another for having completed the preceding process(es).                                                                                                                                                                                    | Simple confirmation<br>Inspection on site                                                                       |
| Pre-flight and Countdown Procedure         | Upon request, the flier must provide the inspector with hardcopy checklist procedures for any of the recovery system's unique final on-pad preparations, pre-flight, and launch (both nominal and off-nominal/abort/mishap flows) - including self-inspection/verification steps which make individual team members accountable to one another for having completed the preceding process(es).                                                                                                                                                                       | Simple confirmation<br>Inspection on site                                                                       |
| Dual-event Parachute and Parafoil Recovery | Each independently recovered launch vehicle body, anticipated to reach an apogee above 450 m above ground level (AGL), shall follow a "dual-event" recovery operations concept, including an initial deployment event (e.g., a drogue parachute deployment; reefed main parachute deployment or similar) and a main deployment event (e.g., a main parachute deployment; main parachute un-reefing or similar). Independently recovered bodies, whose apogee is not anticipated to exceed 450 m AGL, are exempt and may feature only a single/main deployment event. | Proof by calculation<br>Proof by reasoned argumentation                                                         |
| Inspect for Damage                         | If previously flown, any used parachutes, shock chords, and suspension lines must not exhibit signs of damage which threatens the safe recovery of the rocket.                                                                                                                                                                                                                                                                                                                                                                                                       | Simple Confirmation<br>Inspection on site                                                                       |
| Initial Deployment Event                   | The initial deployment event shall occur at or near apogee, stabilize the vehicle's attitude (i.e., prevent or eliminate tumbling), and reduce its descent rate sufficiently to permit the main deployment event, yet not so much as to exacerbate wind drift. Any part, assembly or device, featuring an initial                                                                                                                                                                                                                                                    | Proof by reasoned argument<br>(Deployment event)<br>Proof by calculation<br>(Descent rate)<br>Proof by previous |



|                                     |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                       |
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|                                     | deployment event, shall result in a descent velocity of said item of 23-46 m/s.                                                                                                                                                                                                                                            | testing (Descent rate)                                                                                                                |
| Main Deployment Event               | The main deployment event shall occur at an altitude no higher than 450 m AGL and reduce the vehicle's descent rate sufficiently to prevent excessive damage upon impact with ground. Any part, assembly or device, featuring a main deployment event, shall result in a descent velocity of said item of less than 9 m/s. | Proof by reasoned argumentation (Deployment event)<br>Proof by calculation (Descent rate)<br>Proof by previous testing (Descent rate) |
| Parachutes and Parafoils            | Any parachutes or parafoils used must be rated for the weight of the vehicle and the expected conditions at deployment.                                                                                                                                                                                                    | Proof by calculation                                                                                                                  |
| Safe Descent rate                   | Parachutes or parafoils intended for the final descent phase to the ground must not allow a descent rate that would represent a safety hazard.                                                                                                                                                                             | Proof by calculation<br>Proof by reasoned argumentation<br>Proof by previous testing                                                  |
| Personal Safety                     | The arming/disarming process must not place the operator in the predicted path of hot gases, ejecta, or deployable devices which might result from an unintentional triggering event                                                                                                                                       | Simple check                                                                                                                          |
| Activation Devices                  | The electronics controlling recovery events must be activated by externally accessible switches, and do not require any disassembly of the rocket to either activate or de-activate.                                                                                                                                       | Simple confirmation                                                                                                                   |
| Positive State Indication           | Each independent set of electronics controlling recovering events must provide sensory (i.e., visual or auditory) indication of its activation.                                                                                                                                                                            | Simple confirmation<br>Inspection on site                                                                                             |
| Acceleration Effects on Electronics | Heavy items - most notably batteries - must be adequately supported to prevent them becoming dislodged under anticipated flight loads.                                                                                                                                                                                     | Simple confirmation                                                                                                                   |
| Ejection Gas Protection             | The recovery system shall implement adequate protection (e.g., fire-resistant material, pistons, baffles etc.) to prevent hot ejection gases (if implemented) from causing burn damage to retaining chords, parachutes, and other vital components as the specific design demands.                                         | Simple confirmation<br>Inspection on site                                                                                             |
| Parachute Swivel Links              | The recovery system rigging (e.g., parachute lines, risers, shock chords, etc.) shall implement swivel links at connections to relieve torsion, as the specific design demands. This will mitigate the risk of torque                                                                                                      | Simple confirmation<br>Inspection on site                                                                                             |



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|                                                      | loads unthreading bolted connections during recovery as well as parachute lines twisting up.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                          |
| Parachute Coloration and Markings                    | When separate parachutes are used for the initial and main deployment events, these parachutes should be visually highly dissimilar from one another. This is typically achieved by using parachutes whose primary colours contrast those of the other chute. This will enable ground-based observers to characterize deployment events more easily with high-power optics. Utilised parachutes should use colours providing a clear contrast to a blue sky and a grey/white cloud cover. | Simple confirmation                                                                                      |
| Non-parachute/Parafoil Recovery Systems              | Teams exploring other recovery methods (i.e., non-parachute or parafoil based) shall mention them in the dedicated field of the Technical Questionnaire. The organisers may make additional requests for information and draft unique requirements depending on the team's specific design implementation.                                                                                                                                                                                | Simple confirmation<br>Inspection on site<br>Proof by reasoned argumentation<br>In-depth proofing needed |
| <b>REDUNDANT ELECTRONICS</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                          |
| Redundant COTS Recovery Electronics                  | At least one redundant recovery system electronics subsystem shall implement a COTS flight computer. To be considered COTS, the flight computer (including flight software) must have been developed and validated by a commercial third party.                                                                                                                                                                                                                                           | Simple confirmation                                                                                      |
| Mandatory Official GPS Tracking and Tracking Systems | EuRoC will require teams to implement a common mandatory GPS tracking and locating device in all rocket systems featuring a dual-event deployment and recovery system.                                                                                                                                                                                                                                                                                                                    | Simple confirmation                                                                                      |
| Dissimilar Redundant Recovery Electronics            | There is no requirement that the redundant/backup system be dissimilar to the primary; however, there are advantages to using dissimilar primary and backup systems. Such configurations are less vulnerable to any inherent environmental sensitivities, design, or production flaws affecting a particular component.                                                                                                                                                                   | No action necessary                                                                                      |
| <b>SAFETY CRITICAL WIRING</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                          |



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| Cable Management                  | All safety critical wiring shall implement a cable management solution (e.g., wire ties, wiring, harnesses, cable raceways) which will prevent tangling and excessive free movement of significant wiring/cable lengths due to expected launch loads. This requirement is not intended to negate the small amount of slack necessary at all connections/terminals to prevent unintentional de-mating due to expected launch loads transferred into wiring/cables at physical interfaces.                                                                                                                                                                                      | Simple confirmation<br>Inspection on site |
| Secure Connections                | All safety critical wiring/cable connections shall be sufficiently secure as to prevent de-mating due to expected launch loads. This will be evaluated by a "tug test", in which the connection is gently but firmly "tugged" by hand to verify it is unlikely to break free in flight.                                                                                                                                                                                                                                                                                                                                                                                       | Inspection on site                        |
| Cryo-compatible Wire Insulation   | In case of propellants with a boiling point of less than -50°C any wiring or harness passing within the close proximity of a cryogenic device (e.g., valve, piping, etc.) or a cryogenic tank (e.g., a cable tunnel next to a LOX tank) shall utilize safety critical wiring with cryo-compatible insulation (i.e., Teflon, PTFE, etc.).                                                                                                                                                                                                                                                                                                                                      | Inspection on site                        |
| Recovery System Energetic Devices | All stored-energy devices (aka energetics) used in recovery systems shall comply with the energetic device requirements defined in Section 4. of this document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Simple confirmation                       |
| <b>RECOVERY SYSTEM TESTING</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |
| Ground Test Demonstration         | All recovery system mechanisms shall be successfully (without significant anomalies) tested prior to EuRoC, either by flight testing, or through one or more ground tests of key subsystems. In the case of such ground tests, sensor electronics will be functionally included in the demonstration by simulating the environmental conditions under which their deployment function is triggered. The test results and a statement of a successful test, complete with dates and signatures (at least three) are considered a mandatory deliverable and annex to the Technical Report. Failure to deliver this annex will automatically result in a "denied" flight status. | Proof by previous testing                 |



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| Optional Flight Test Demonstration | All recovery system mechanisms shall be successfully (without significant anomalies) tested prior to EuRoC, either by flight testing, or through one or more ground tests of key subsystems. While not required, a flight test demonstration may be used in place of ground testing. In the case of such a flight test, the recovery system flown will verify the intended design by implementing the same major subsystem components (e.g., flight computers and parachutes) as will be integrated into the launch vehicle intended for EuRoC (i.e., a surrogate booster may be used). The test results and a statement of a successful test, complete with dates and signatures (at least three) are considered a mandatory deliverable and annex to the Technical Report. Failure to deliver this annex will automatically result in a “denied” flight status. | No action necessary                       |
| <b>STORED-ENERGY DEVICES</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |
| Energetic Device Safing and Arming | All energetics shall be “safed” until the rocket is in the launch position, at which point they may be “armed”. An energetic device is considered safed when two separate events are necessary to release the energy of the system. An energetic device is considered armed when only one event is necessary to release the energy. For the purpose of this document, energetics are defined as all stored-energy devices – other than propulsion systems – that have reasonable potential to cause bodily injury upon energy release. See Section 3. for more information.                                                                                                                                                                                                                                                                                       | Simple check                              |
| Arming Device Access               | All energetic device arming features shall be externally accessible/controllable. This does not preclude the limited use of access panels which may be secured for flight while the vehicle is in the launch position.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Simple confirmation<br>Inspection on site |
| Arming Device Location             | All energetic device arming features shall be located on the airframe such that any inadvertent energy release by these devices will not impact personnel arming them. For example, the arming key switch for an energetic device used to deploy a hatch panel shall not be located at the same airframe clocking position as the hatch panel deployed by that charge. Furthermore, it is highly recommended that the arming mechanism is accessible from ground level, without the use of ladders or other elevation                                                                                                                                                                                                                                                                                                                                             | Simple confirmation                       |



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|                                                        | devices, when the rocket is at a vertical orientation on the launch rail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                     |
| <b>SRAD PRESSURE VESSELS</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |
| Relief Device                                          | SRAD pressure vessels shall implement a relief device, set to open at no greater than the proof pressure specified in the following requirements. SRAD (including modified COTS) rocket motor propulsion system combustion chambers are exempted from this requirement.                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Proof by previous testing                                                           |
| Designed Burst Pressure for Metallic Pressure Vessels  | SRAD and modified COTS pressure vessels constructed entirely from isentropic materials (e.g., metals) shall be designed to a burst pressure no less than 2 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.                                                                                                                                                                                                                                                                                                                                                                                     | Proof by calculation<br>Proof by reasoned argumentation<br>In-depth proofing needed |
| Designed Burst Pressure for Composite Pressure Vessels | All SRAD and modified COTS pressure vessels either constructed entirely from non-isentropic materials (e.g., fibre reinforced plastics; FRP; composites) or implementing composite overwrap of a metallic vessel (i.e., composite overwrapped pressure vessels; COPV), shall be designed to a burst pressure no less than 3 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.                                                                                                                                                                                                                    | Proof by calculation<br>Proof by reasoned argumentation<br>In-depth proofing needed |
| <b>SRAD PRESSURE VESSEL TESTING</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |
| Proof Pressure Testing                                 | SRAD and modified COTS pressure vessels shall be proof pressure tested successfully (without significant anomalies) to 1.5 times the maximum expected operating pressure for no less than twice the maximum expected system working time, using the intended flight article(s) (e.g., the pressure vessel(s) used in proof testing must be the same one(s) flown at EuRoC). The maximum system working time is defined as the maximum uninterrupted time duration the vessel will remain pressurized during pre-launch, flight, and recovery operations.<br>The test results and a statement of a successful test, complete with dates and signatures (at least three) are considered mandatory deliverable and annexed | Proof by previous testing                                                           |



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|                                      | to the Technical Report. Failure to deliver this annex will automatically result in a “denied” flight status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                       |
| Optional Burst Pressure Testing      | Although there is no requirement for burst pressure testing, a rigorous verification & validation test plan typically includes a series of both non-destructive (i.e., proof pressure) and destructive (i.e., burst pressure) tests. A series of burst pressure tests performed on the intended design will be viewed favourably; however, this will not be considered an alternative to proof pressure testing of the intended flight article.                                                                                                                                                                                                                                                | No action necessary                                   |
| <b>ACTIVE FLIGHT CONTROL SYSTEMS</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |
| Restricted Control Functionality     | Launch vehicle active flight control systems shall be optionally implemented strictly for pitch and/or roll stability augmentation, or for aerodynamic "braking". Under no circumstances will a launch vehicle entered in EuRoC be actively guided towards a designated spatial target. The organisers may make additional requests for information and draft unique requirements depending on the team's specific design implementation.                                                                                                                                                                                                                                                      | Simple confirmation                                   |
| Unnecessary for Stable Flight        | Launch vehicles implementing active flight controls shall be naturally stable without these controls being implemented (e.g., the launch vehicle may be flown with the control actuator system [CAS] — including any control surfaces — either removed or rendered inert and mechanically locked, without becoming unstable during ascent). Attitude Control Systems (ACS) will serve only to mitigate the small perturbations which affect the trajectory of a stable rocket that implements only fixed aerodynamic surfaces for stability. The organisers may make additional requests for information and draft unique requirements depending on the team's specific design implementation. | Proof by reasoned argumentation<br>Inspection on site |
| Designed to Fail Safe                | Control Actuator Systems (CAS) shall mechanically lock in a neutral state whenever either an abort signal is received for any reason, primary system power is lost, or the launch vehicle's attitude exceeds 30° from its launch elevation. Any one of these conditions being met will trigger the fail-safe, neutral system state. A neutral state is defined as one which does not apply any moments to the                                                                                                                                                                                                                                                                                  | Proof by reasoned argumentation<br>Inspection on site |



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|                                          | launch vehicle (e.g., aerodynamic surfaces trimmed or retracted, gas jets off, etc.).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                       |
| Boost Phase Dormancy                     | CAS shall mechanically lock in a neutral state until either the mission's boost phase has ended (i.e., all propulsive stages have ceased producing thrust), the launch vehicle has crossed the point of maximum aerodynamic pressure (i.e., max Q) in its trajectory, or the launch vehicle has reached an altitude of 6.000 m AGL. Any one of these conditions being met will permit the active system state. A neutral state is defined as one which does not apply any moments to the launch vehicle (e.g., aerodynamic surfaces trimmed or retracted, gas jets off, etc.). | Proof by reasoned argumentation<br>Inspection on site |
| Active Flight Control System Electronics | Wherever possible, all active control systems should comply with requirements and goals for "redundant electronics" and "safety critical wiring" as recovery systems — understanding that in this case "initiation" refers CAS commanding rather than a recovery event. Flight control systems are exempt from the requirement for COTS redundancy, given that such components are generally unavailable as COTS to the amateur high-power rocketry community.                                                                                                                 | Simple confirmation                                   |
| Active Flight Control System Energetics  | All stored-energy devices used in an active flight control system (i.e., energetics) shall comply with the energetic device requirements defined in Section 4. of this document.                                                                                                                                                                                                                                                                                                                                                                                               | Simple confirmation                                   |
| <b>AIRFRAME STRUCTURES</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |
| Adequate Venting                         | Launch vehicles shall be adequately vented to prevent unintended internal pressures developed during flight from causing either damage to the airframe or any other unplanned configuration changes. Typically, a 3 mm to 5 mm hole is drilled in the booster section just behind the nosecone or payload shoulder area, and through the hull or bulkhead of any similarly isolated compartment/bay.                                                                                                                                                                           | Simple confirmation<br>Inspection on site             |
| <b>OVERALL STRUCTURAL INTEGRITY</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                       |
| Checklist                                | Upon request, the flier should provide the inspector with hardcopy checklist procedures for the rocket's assembly and integration for flight - including self-inspection/verification steps which make individual                                                                                                                                                                                                                                                                                                                                                              | Simple confirmation<br>Inspection on site             |



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|                                   | team members accountable to one another for having completed the preceding process(es).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                |
| Material Selection                | PVC (and similar low-temperature polymers), Public Missiles Ltd. (PML) Quantum Tube components shall not be used in any structural (i.e., load bearing) capacity, most notably as load bearing eyebolts, launch vehicle airframes, or propulsion system combustion chambers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No action necessary<br>(for stainless steel components)<br>Simple confirmation |
| Load Bearing Eyebolts and U-bolts | All load bearing eyebolts shall be of the closed-eye, forged type — NOT of the open eye, bent wire type. Furthermore, all load bearing eyebolts and U-Bolts shall be steel or stainless steel. This requirement extends to any bolt and eye-nut assembly used in place of an eyebolt.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No action necessary<br>(for stainless steel)<br>Inspection on site             |
| Implementing Coupling Tubes       | Airframe joints which implement "coupling tubes" should be designed such that the coupling tube extends no less than one body calibre on either side of the joint — measured from the separation plane. Regardless of implementation (e.g., RADAX or other join types) airframe joints will be "stiff" (i.e., prevent bending).                                                                                                                                                                                                                                                                                                                                                                                                                              | Simple confirmation<br>Proof by reasoned argumentation                         |
| Launch Lug Mechanical Attachment  | Launch lugs (i.e., rail guides) should implement "hard points" for mechanical attachment to the launch vehicle airframe. These hardened/reinforced areas on the vehicle airframe, such as a block of wood installed on the airframe interior surface where each launch lug attaches, will assist in mitigating lug "tear outs" during operations. The aft most launch lug shall support the launch vehicle's fully loaded launch weight while vertical. At EuRoC, competition officials will require teams to lift their launch vehicles by the rail guides and/or demonstrate that the bottom guide can hold the vehicle's weight when vertical. This test needs to be completed successfully before the admittance of the team to Launch Readiness Review. | Inspection on site<br>Proof by previous testing                                |
| Launch Rail Fit Check             | All teams shall perform a "launch rail fit check" as a part of the flight preparations (the Launch Readiness Review), before going to the launch range. This requirement is particularly important if a team is not bringing their own launch rail, but instead relying on EuRoC provided launch rails.                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Inspection on site                                                             |



|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |
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| Rail Guide Attachment  | The rail guides must be firmly attached to the rocket without evidence of cracking in the joints, and the aft most guide attachment must be sufficient to bear the rocket's entire mass when erected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Inspection on site              |
| Slip-fit Joints        | Joints intended to separate in flight cannot become separated when loaded by their own weight alone, and the Flier should demonstrate cognizance of shear pin design (if implemented).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Proof by reasoned argumentation |
| Joint Stiffness        | All joints - both separating and non-separating in flight - must be "stiff", so as to eliminate any visible airframe bending.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Inspection on site              |
| Fin Attachment         | The fins must be firmly attached to the rocket without evidence of cracking in the joints. ("Hairline" cracks may be acceptable if the fins are not loose or, if the fins are mounted using the "through-the-wall" [TTW] construction technique.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Inspection on site              |
| Fin Stiffness          | The fins must exhibit no shifting and minimal deflection (i.e., bending) when handled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Inspection on site              |
| Fin "Warping"          | The fins must exhibit little-to-no indication of damage due to moisture penetration or excessive thermal cycling during storage or transport - leading to out of tolerance dimensional changes in the part.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Inspection on site              |
| <b>RF TRANSPARENCY</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |
| RF Window Location     | <p>Any internally mounted RF transmitter, receiver or transceiver, not having the applicable antenna or antennas mounted externally on the airframe, shall employ "RF windows" in the airframe shell plating (typically glass fibre panels), enabling RF devices with antennas mounted inside the airframe, to transmit the signal through the airframe shell. RF windows in the flight vehicle shell shall be a 360° circumference and be at least two body diameters in length. The internally mounted RF antenna(s) shall be placed at the midpoint of the RF window section, facilitating maximizing the azimuth radiation pattern.</p> <p>RF transmitter, receivers or transceivers are not allowed to be mounted externally. Externally mounted antennas are allowed, but only if at least two antennas are mounted on opposite sides of the airframe, thus retaining circumferential symmetry and covering sufficient transmission area, transmitting or receiving identical signals.</p> | Simple confirmation             |



|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |
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|                      | As popular as carbon fibre is for the construction of strong and lightweight airframes, it is also conductive and will significantly shield and/or degrade RF signals, which is unacceptable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |
| Identifying Markings | The team's Team ID (a number assigned by EuRoC prior to the competition event), project name, and academic affiliation(s) shall be clearly identified on the launch vehicle airframe. The Team ID especially, will be prominently displayed (preferably visible on all four quadrants of the vehicle, as well as fore and aft), assisting competition officials to positively identify the project hardware with its respective team throughout EuRoC.                                                                                                                                                                                                                          | No action necessary |
| Other Markings       | There are no requirements for airframe coloration or markings beyond those specified in Section 6.4. of this document. However, EuRoC offers the following recommendations to student teams: mostly white or lighter tinted colour (e.g., yellow, red, orange, etc.) airframes are especially conducive to mitigating some of the solar heating experienced in the EuRoC launch environment. Furthermore, high-visibility schemes (e.g., high-contrast black, orange, red, etc.) and roll patterns (e.g., contrasting stripes, "V" or "Z" marks, etc.) may allow ground-based observers to more easily track and record the launch vehicle's trajectory with high-power optics. | No action necessary |
| <b>PAYLOAD</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |
| Payload Mass         | The launch vehicle shall carry no less than 1000 g of payload. Teams are responsible for conducting a "weigh-in" on site in the presence of the competition officials. Competition officials will accept payload weigh-ins as much as 5% (50 g) less than the specified minimum. If this requirement is not met, "nominal" flight status for the payload may be denied by the officials during FRR, resulting in an action item to increase payload mass.                                                                                                                                                                                                                       | Inspection on site  |



|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                               |
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| Payload recovery                                               | <p>Payloads may be deployable or remain attached to the launch vehicle throughout the flight. Deployable payloads shall incorporate an independent recovery system, reducing the payload's descent velocity to less than 9 m/s before it descends through an altitude of 450 m AGL. Deployable payloads without two-stage recovery systems (drogue and main chute, like the rockets) will be subjective to considerable drift during descent.</p> | <p>Proof by calculation<br/>Proof by reasoned argumentation<br/>Proof by previous testing</p> |
| Payload Recovery System Electronics and Safety Critical Wiring | <p>Payloads implementing independent recovery systems shall comply with the same requirements and goals as the launch vehicle for "redundant electronics" and "safety critical wiring".</p>                                                                                                                                                                                                                                                       | <p>Inspection on site</p>                                                                     |
| Payload Recovery System Testing                                | <p>Payloads implementing independent recovery systems shall comply with the same requirements and goals as the launch vehicle for "recovery system testing".</p>                                                                                                                                                                                                                                                                                  | <p>Simple confirmation</p>                                                                    |
| Deployable Payload Landing Site Tracking Required              | <p>It must be noted that deployable payloads are equivalent to flight vehicle bodies and sections, in that they can be difficult to locate after landing. All deployable payloads shall feature the same mandatory landing site tracking system as all rockets and rocket stages as specified in Section 3.5. The landing site locator ID must comply with the call-sign requirements defined in Sections 3.5.3. and 3.5.4.</p>                   | <p>Simple confirmation</p>                                                                    |
| Payload Energetic Devices                                      | <p>All stored-energy devices (i.e., energetics) used in payload systems shall comply with the energetic device requirements defined in Section 4. of this document.</p>                                                                                                                                                                                                                                                                           | <p>Simple confirmation</p>                                                                    |
| <b>LAUNCH AND ASCENT TRAJECTORY REQUIREMENTS</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                               |
| Launch Azimuth and Elevation                                   | <p>Launch vehicles shall nominally launch at an elevation angle of <math>84^{\circ} \pm 1^{\circ}</math> and a launch azimuth defined by competition officials at EuRoC. Competition officials reserve the right to require certain vehicles' launch elevation be as low as <math>70^{\circ}</math>, if flight safety issues are identified during pre-launch activities.</p>                                                                     | <p>Simple check</p>                                                                           |



|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Launch Stability                              | Launch vehicles shall have sufficient velocity upon "departing the launch rail" to ensure they will follow predictable flight paths. In lieu of detailed analysis, a rail departure velocity of at least 30 m/s is generally acceptable. Alternatively, the team may use detailed analysis to prove stability is achieved at a lower rail departure velocity 20 m/s either theoretically (e.g., computer simulation) or empirically (e.g., flight testing).                    | Proof by calculation     |
| Ascent Stability                              | Launch vehicles shall remain "stable" for the entire ascent. Stable is defined as maintaining a static margin of at least 1.5 to 2 body calibres, regardless of CG movement due to depleting consumables and shifting centre of pressure (CP) location due to wave drag effects (which may become significant as low as 0.5 Mach). Not falling below 2 body calibres will be considered nominal, while falling below 1.5 body calibres will be considered a loss of stability. | Proof by calculation     |
| Over-stability                                | All launch vehicles should avoid becoming "over-stable" during their ascent. A launch vehicle may be considered over-stable with a static margin significantly greater than 2 body calibres (e.g., greater than 6 body calibres).                                                                                                                                                                                                                                              | Proof by calculation     |
| Flight Simulation                             | Upon request, the flier should either provide a hard copy, or demonstrate on a portable computer, a 3-degrees-of-freedom (3DoF) simulation (or better) of the rocket's nominal trajectory.                                                                                                                                                                                                                                                                                     | In-depth proofing needed |
| Fin Alignment                                 | The fins should be mounted parallel to the roll axis of the rocket, or (if canted or otherwise roll inducing) the Flier must demonstrate cognizance of the predicted roll behaviour and its effects.                                                                                                                                                                                                                                                                           | Inspection on site       |
| Staging Event Sequence and Timing             | Any delays implemented between staging events must not be so long as to significantly risk the rocket having "arced-over" into an unsafe orientation - typically by "gravity turn".                                                                                                                                                                                                                                                                                            | Proof by calculation     |
| <b>TEAM-PROVIDED LAUNCH SUPPORT EQUIPMENT</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |
| Equipment Portability                         | If possible/practicable, teams should make their launch support equipment man-portable over a short distance (a few hundred metres). Environmental considerations at the launch site permit only limited vehicle use beyond designated roadways, campgrounds, and basecamp areas.                                                                                                                                                                                              | Simple confirmation      |



|                               |                                                                                                                                                                                                                                                                                                                                               |                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Launch Rail Elevation         | Team provided launch rails shall implement the nominal launch elevation specified in Section 7. of this document and, if adjustable, not permit launch at angles either greater than the nominal elevation or lower than 70°.                                                                                                                 | Inspection on site  |
| Operational Range             | All team provided launch control systems shall be electronically operated and have a maximum operational range of no less than 650 metres from the launch rail. The maximum operational range is defined as the range at which launch may be commanded reliably.                                                                              | No action necessary |
| Fault Tolerance and Arming    | All team provided launch control systems shall be at least single fault tolerant by implementing a removable safety interlock (i.e., a jumper or key to be kept in possession of the arming crew during arming) in series with the launch switch.                                                                                             | Inspection on site  |
| Safety Critical Switches      | All team provided launch control systems shall implement ignition switches of the momentary, normally open (also known as "dead man") type so that they will remove the signal when released. Mercury or "pressure roller" switches are not permitted anywhere in team provided launch control systems.                                       | Simple confirmation |
| <b>EQUIPMENT</b>              |                                                                                                                                                                                                                                                                                                                                               |                     |
| Communication Equipment       | All teams must bring any Personal Protection Equipment (PPE) required for all preparation- and launch activities. EuRoC does not have a supply of spare PPE. PPE includes, but is not limited to, safety goggles, gloves, safety shoes, hardhats, ear protection, cryo-protection, etc.                                                       | No action necessary |
| Personal Protection Equipment | All teams must bring any Personal Protection Equipment (PPE) required for all preparation- and launch activities. EuRoC does not have a supply of spare PPE. PPE includes, but is not limited to, safety goggles, gloves, safety shoes, hardhats, ear protection, cryo-protection, etc.                                                       | Simple confirmation |
| Field Equipment               | All teams are encouraged to provide each participating team member with a suitable "field/day pack", which is kept close at hand (or worn) during launch days. Due to the possibility of strong sunlight and high temperatures even in October, some of these provisions are intended to get students through a hot and dry day in the field, | No action necessary |



|  |                                                                                                                                                                                                               |  |
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|  | while other provisions are intended to enable student teams to continue efficient operation after loss of daylight after a quick sun-down and a resulting sudden and significant drop in ambient temperature. |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

Table 9: Legend for de-scoping FRR checklist.

| LEGEND FOR DE-SCOPING FEEDBACK           |
|------------------------------------------|
| This requirement is very important       |
| This requirement is important            |
| This requirement is of lesser importance |

Table 10: Legend for actions to be taken on the FRR checklist.

| ACTIONS TO BE TAKEN             |                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No action necessary             | "I see you used stainless steel here. Okay, fine"                                                                                                                                                    |
| Simple confirmation             | "Are you using non-toxic propellants?" – "Yes, we are"                                                                                                                                               |
| Simple check                    | "Is everybody at least 15 m away when the ground-start ignition circuit is arming?" – "Okay now, yes"                                                                                                |
| Inspection on site              | "Are all the critical wiring/cable connections sufficiently secured?" – "I will have a look, ah I see, yes"                                                                                          |
| Proof by reasoned argumentation | "Can you tell me about your process of offloading propellant in case of a launch abort?" – "Okay, sounds reasonable, this should work."                                                              |
| Proof by previous testing       | "Have you tested the pressure vessels to 1.5 the maximum expected operating pressure?" – "Okay, I will have a look at the results and understand if everything has been tested appropriately."       |
| Proof by calculation            | "Regarding the launch stability, have you calculated the lower rail departure velocity? How did you do it? What is the result?" – "Okay, I see and understand the calculation, this will work then." |
| In-depth proofing needed        | "How does this design feature work?" – "Okay, so you are not certain, and I do not understand on site, so let us go to the CAD model and check."                                                     |